Hints: Abstract algebra – 1. Finite groups: counting

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. In the commutative case, raise ab to the power rs: $(ab)^{rs} = a^{rs}b^{rs} = e$. Together with $(ab)^t = e$ and $\gcd(rs, t) = 1$ this gives $ab = e$.

Hint 2. So $b = a^{-1}$, hence $a^r = e$ and $a^s = b^{-s} = e$. Since $\gcd(r, s) = 1$, $a = e$.

Hint 3. In general the answer is no. Look for a of order 2 and b of order 3 with ab of order 5, e.g. in A_5: $a = (12)(34)$, $b = (135)$ (check that ab is a 5-cycle). Then $r = 2$, $s = 3$, $t = 5$ work.

Problem 2

Hint 1. Every $x \neq e$ commutes with its powers. If $\operatorname{ord}(x) \geqslant 4$, then x, x^2, x^{-1} give two different commuting pairs. So all non-identity elements have order 2 or 3. If there were two different subgroups of order 3, they would give two pairs.

Hint 2. If no element has order 3, then $x^2 = e$ for all x, G is abelian, and the number of commuting pairs is never exactly one. So there is a unique subgroup $C = \{e, c, c^2\}$ of order 3, and $\{c, c^2\}$ is the only commuting pair. Hence $C(c) = C$, and C is normal.

Hint 3. Conjugation gives a homomorphism $G \to \operatorname{Aut}(C) \cong \mathbb{Z}/2$ with kernel $C(c) = C$. So $|G| \in \{3, 6\}$. A group of order 6 in which c commutes with no involution is non-abelian, hence S_3.

Problem 3

(a)

Hint 1. $|H \cap xH| = |H| + |xH| - |H \cup xH| \geqslant 2p - n$.

(b)

Hint 1. Take $x \in H$. If $y = xh \in H \cap xH$ with $h \in H$, then x, h, xh are all involutions (or e). Show that this forces $xh = hx$.

Hint 2. So the centraliser $C(x)$ contains the $\geqslant 2p - n > n/2$ elements $h \in H$ with $xh \in H$. A subgroup with more than $n/2$ elements is all of G, so every $x \in H$ is central.

Hint 3. The subgroup generated by H has more than $n/2$ elements, so it is G. A group generated by central elements is abelian.

(c)

Hint 1. If G were abelian, H would be a subgroup. With $|H| > n/2$ this forces $H = G$, i.e. $p = n > 3n/4$, a contradiction.

Problem 4

Hint 1. Both statements are true. For (b), try groups with very few subgroups.

Hint 2. (b) $\mathbb{Z}/p$ with p prime has exactly 2 subgroups, so $n/s = p/2$. (a) Take $G = (\mathbb{Z}/2)^k$, an $\mathbb{F}_2$-vector space: its subgroups are its subspaces. Count the subspaces of dimension $\lfloor k/2 \rfloor$, or just bound their number from below by about $2^{k^2/4}$, which is much larger than 2^k.

Problem 5

Note. The "only if" part fails for $n = 1$, since then all $H^x = \{e\}$. The hints assume $n \geqslant 2$; the formula for the union holds in all cases.

Hint 1. If $x^{-1}y \in H$, then $H^y = H^{x(x^{-1}y)} = H^x$. Conversely, $H^x = H^y$ gives $H^{x^{-1}y} = H$, and by hypothesis this is impossible if $x^{-1}y \notin H$ (it would force $H = \{e\}$).

Hint 2. So the distinct conjugates correspond to left cosets of H: there are m/n of them. Two distinct ones meet only in e, since $H^x \cap H^y = (H \cap H^{x^{-1}y})^x$.

Hint 3. Hence $\left|\bigcup_x H^x\right| = \frac{m}{n}(n-1) + 1 = m - \frac{m}{n} + 1$.

Problem 6

Hint 1. First produce a neutral element. Choose e with $c * e = c$, and show that $w * e = w$ for every w: write $w = c * u$, using the hypothesis with $x = c$.

Hint 2. Then pick c' with $c * c' = e$ and multiply $a * c = b * c$ by c', using associativity and commutativity.

Problem 7

Hint 1. If $G/Z(G) = \langle gZ \rangle$, write $x = g^i z_1$, $y = g^j z_2$ with z_1, z_2 central and multiply out.

Hint 2. For $|G| = p^2$: the class equation gives $Z(G) \neq \{e\}$, so $G/Z(G)$ has order 1 or p and is cyclic.

Hint 3. For order p^3, take the upper unitriangular 3×3 matrices over $\mathbb{F}_p$ (the Heisenberg group). For $p = 2$ this is the dihedral group of order 8.

Problem 8

Hint 1. Let $K = \ker \pi$ and take $H \leqslant G$ *minimal* with $\pi(H) = Q$. Note $|H| = |H \cap K| \cdot |Q|$. Suppose a prime r divides $|H|$ but not $|Q|$, and let P be a Sylow r-subgroup of $H \cap K$. Then P is even a Sylow r-subgroup of H.

Hint 2. Frattini argument: $H \cap K \trianglelefteq H$, and all Sylow r-subgroups of $H \cap K$ are conjugate inside $H \cap K$, so $H = (H \cap K)\, N_H(P)$. Hence $\pi(N_H(P)) = Q$, and minimality gives $N_H(P) = H$, i.e. $P \trianglelefteq H$.

Hint 3. A normal Sylow subgroup has a complement C, with $H = PC$ and $|C| = |H|/|P|$ (Schur–Zassenhaus; this is the one tool beyond Sylow). Then $\pi(C) = \pi(PC) = Q$ because $P \subseteq K$, and $C < H$, which contradicts minimality.

Problem 9

(a)

Hint 1. Let x have order d. Show that $C(x) = \langle x \rangle$. If not, some prime q divides $|C(x)|/d$, and $q \nmid d$ because n is squarefree. Use Cauchy in $C(x)$ to get an element y of order q commuting with x, and look at the order of xy.

Hint 2. So the conjugacy class of every element of order d has exactly n/d elements, and the set of elements of order d is a union of such classes.

(b)

Hint 1. By (a), $N \geqslant n/d$. The generators of $\langle x \rangle$ give $N \geqslant \varphi(d)$. Multiply, and use that n/d and d are coprime: $N^2 \geqslant \frac{n}{d}\varphi(d) \geqslant \varphi(n/d)\varphi(d) = \varphi(n)$.

Problem 10

Hint 1. Only inner automorphisms are needed. For each g there is $m = m_g$ with $gxg^{-1} = x^m$ for all x. Apply this to $x = g$: you get $\mathrm{ord}(g) \mid m_g - 1$.

Hint 2. Then $[g, y] = gyg^{-1}y^{-1} = y^{m_g - 1}$ lies in $\langle y^{\mathrm{ord}(g)} \rangle$. By symmetry $[y, g] = [g, y]^{-1}$ lies in $\langle g^{\mathrm{ord}(y)} \rangle$.

Hint 3. Let $a = \mathrm{ord}(g)$, $b = \mathrm{ord}(y)$, $\delta = \gcd(a, b)$. The order of $[g, y]$ divides both b/δ and a/δ, which are coprime. Hence $[g, y] = e$.